

INHIBITORS OF PROTEIN SYNTHESIS

1. Aminoglycosides (Gentamicin , streptomycin , Tobramycin, Amikacin)
2. Macrolides (Erythromycin, Clarithromycin, Azithromycin)
3. Lincosamides (Clindamycin)
4. Tetracycline (Doxycycline, Minocycline, Tetracycline)
5. Phenicols (Chloramphenicol)

Aminoglycosides (mycin)

Streptomycin, kanamycin , tobramycin, neomycin, gentamicin, sisomicin, netilmicin, Amikacin.

Common properties of aminoglycosides

Antibacterial activity:

Bactericidal antibiotics with special properties as follows:

- Concentration dependent bactericidal effect
- only acting aerobes

Gram-negative aerobic rods, *E. coli*, *Pseud. aeruginosa*, *Protues*, *Klebsiella*, *Eterobacter*, *Shigella*, *Citrobacter*.

- Aminoglycosides and β -lactams are synergistic and nearly always used in combination with a β -lactam to extend coverage to possibly gram-positive microbes.

Mechanism of action

- What if you have been studying anatomy and physiology all wrong | Kenhub

Protein synthesis (50S inhibitors)

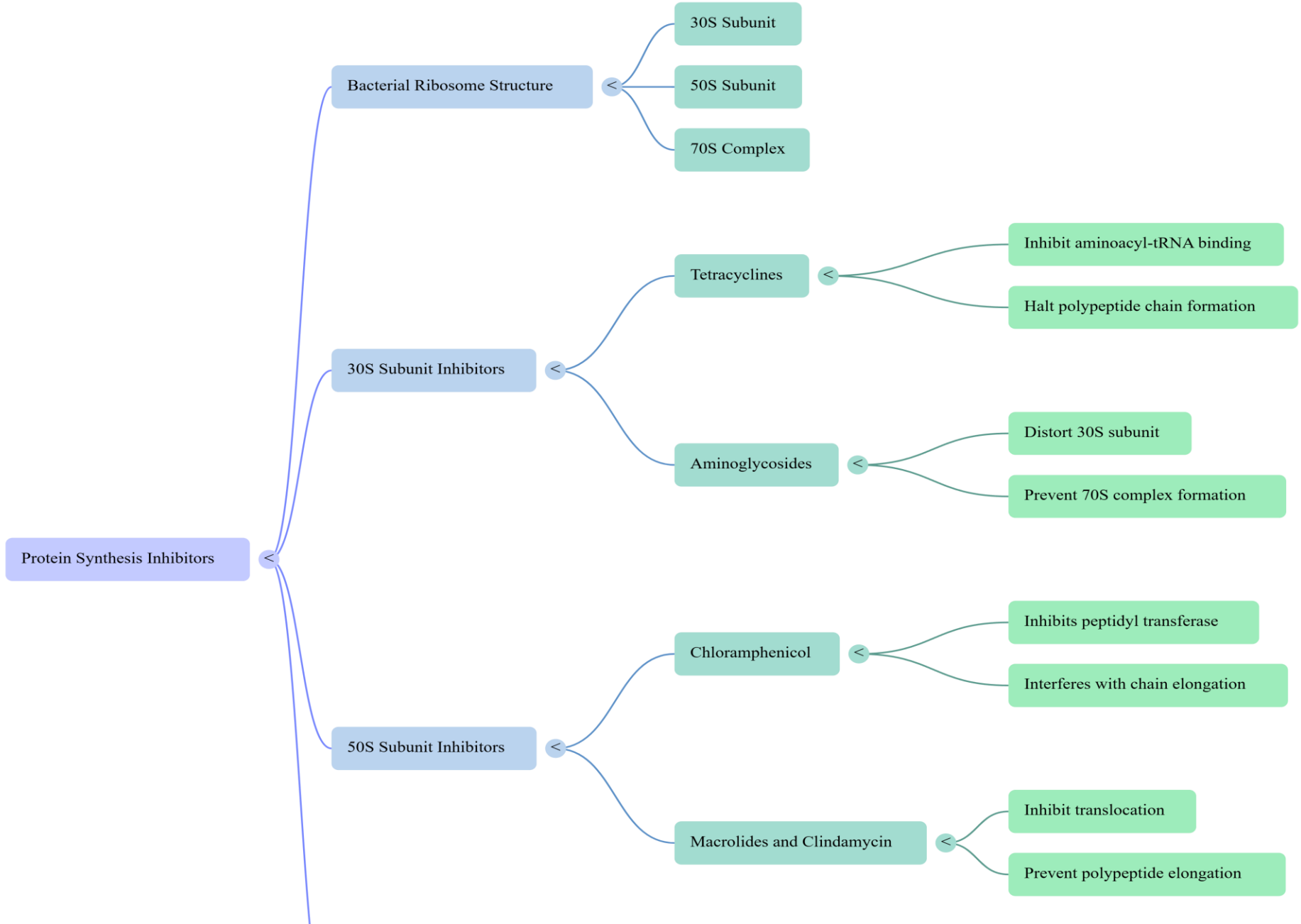
{ Erythromycin (Macrolides)
Chloramphenicol
Clindamycin

Protein synthesis (30S Inhibitors)

{ Tetracycline
Spectinomycin

{ Streptomycin
Gentamicin, Tobramycin
(aminoglycosides)
Amikacin

- Mechanisms:



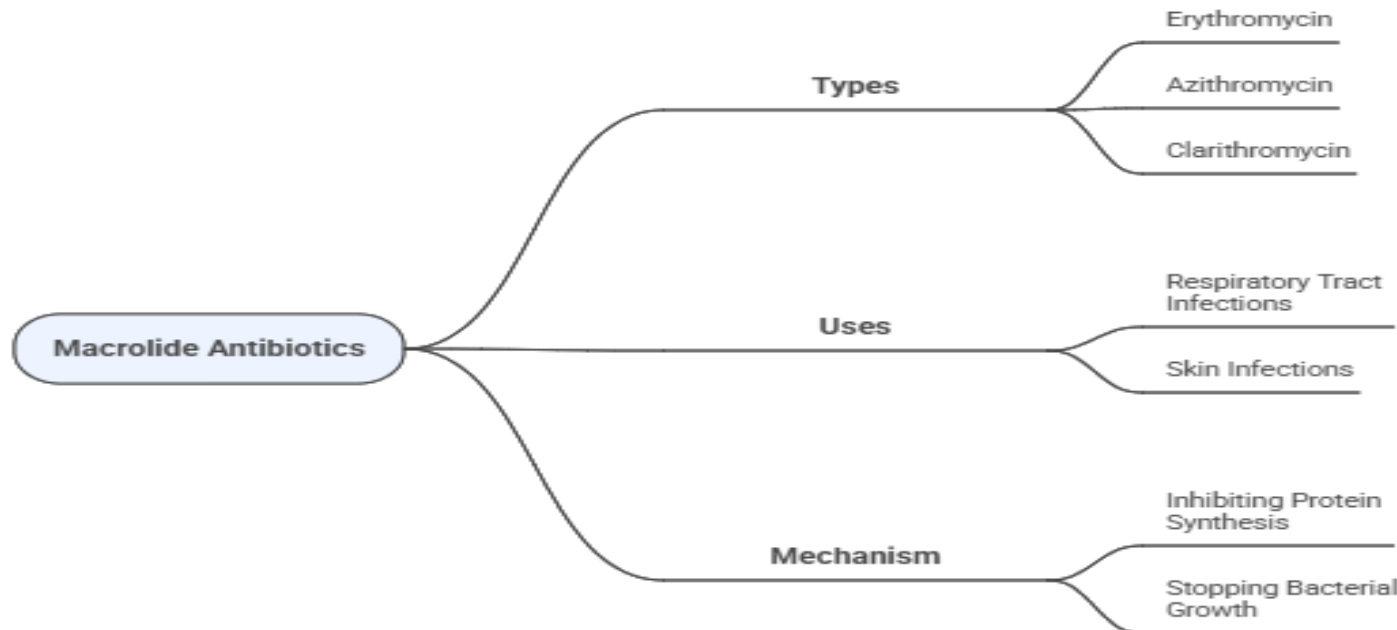
Pharmacokinetics of protein synthesis inhibitors

Drug Class	Absorption (Oral)	Distribution (Vd)	Metabolism	Excretion
Aminoglycosides (e.g., Gentamicin)	Very Poor (0%); requires IV/IM for systemic use.	Limited to extracellular fluid; poor CNS entry.	Not metabolized.	Renal (unchanged via glomerular filtration).
Tetracyclines (e.g., Doxycycline)	60–95%; inhibited by dairy/antacids (chelation).	High; binds to bones and teeth ; poor CNS entry.	Hepatic (varying degrees).	Mixed (Bile/Feces & Urine).
Macrolides (e.g., Azithromycin)	Good; but Erythromycin is acid-labile (needs coating).	Very High; concentrates inside WBCs/tissues .	Hepatic (CYP3A4 inhibitors).	Primarily Biliary (Feces).
Lincosamides (e.g., Clindamycin)	Excellent (90%); not affected by food.	Good; reaches bone and most fluids (no CNS).	Extensive Hepatic metabolism.	Primarily Biliary .
Chloramphenicol	Excellent (90%+).	Widespread; Excellent CNS/CSF penetration.	Hepatic (Glucuronidation).	Renal (as inactive metabolites).

- Adverse reactions:
- Ototoxicity: both vestibular and auditory damage may occur.
- Nephrotoxicity: dose-related, usually reversible.
- Neuromuscular blockade
- Allergic reactions: rashes, fever and angioedema.

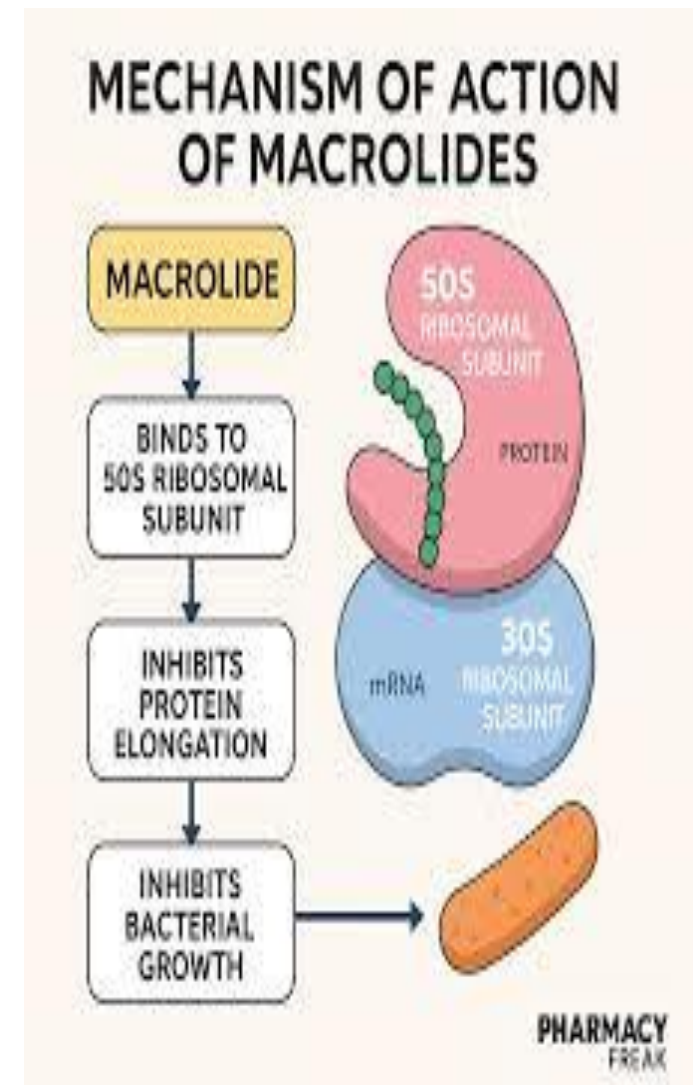
Macrolide antibiotic

Macrolide Antibiotics: Types, Uses, and Mechanism



Mechanism of action

- Inhibits protein synthesis by reversibly binding to the 50S ribosomal subunit
- generally bacteriostatic (may be –cidal
- depending on the concentration and on the type of microorganisms).



Erythromycin

- **Penicillin Alternative:** Historically the first-choice substitute for treating acute orofacial infections in patients with a penicillin allergy.
- Effective against Gram-positive cocci and dental pathogens, **though its use is declining due to frequent GI upset (nausea/cramping).**
- **A potent CYP3A4 inhibitor** that can dangerously increase blood levels of drugs like Warfarin, Carbamazepine, or Theophylline.
- Unlike newer macrolides (like Azithromycin), it is acid-labile and requires **frequent dosing (4 times daily) to maintain therapeutic levels.**

Adverse Reactions of Macrolides

- Gastrointestinal (GI) and other reactions:
 - Nausea
 - Vomiting
 - Diarrhea
 - Abdominal pain or cramping
 - Pseudomembranous colitis
 - Visual disturbances

Macrolides: Contraindications and Precautions

- **Contraindicated** in patients with a **hypersensitivity** to the macrolides; pre-existing liver disease.
- Used **cautiously** in patients with liver dysfunction; myasthenia gravis; during pregnancy or lactation (pregnancy category B and C)

Category	Meaning
A	Controlled studies show no risk to fetus
B	No evidence of risk in humans (animal studies safe)
C	Risk cannot be ruled out
D	Positive evidence of fetal risk , but may be used if benefit outweighs risk
X	Contraindicated in pregnancy (fetal abnormalities proven)

Lincosamides

Clindamycin are bacteriostatic antibiotics that inhibit protein synthesis by binding to the 50S ribosomal subunit.

- They are highly effective against Gram-positive cocci and anaerobes, which are the primary pathogens found in oral infections.
- In dentistry, they serve as a potent alternative for patients with Penicillin allergies due to their excellent bone penetration.

Uses in Dental Practice

1. Treating acute dentoalveolar abscesses and deep jaw infections.
2. Serving as a first-line alternative for patients allergic to Beta-lactams.
3. Managing aggressive gum disease that doesn't respond to standard scaling.
4. Prophylaxis: Preventing Infective Endocarditis or prosthetic joint infection before invasive procedures.
5. Treating bone-centered infections due to their superior ability to reach therapeutic levels in mandibular and maxillary bone.

TETRACYCLINES

Chlortetracycline, oxytetracycline, tetracycline, demeclocycline, methacycline, doxycycline, minocycline

- Tetracyclines are broad-spectrum, bacteriostatic protein synthesis inhibitors that bind to the **30S ribosomal subunit**.
- They are highly effective against both Gram-positive and Gram-negative bacteria, though their use is limited by widespread resistance.

Chloramphenicol



- Broad-spectrum antibiotic that inhibits bacterial protein synthesis by binding to the **50S ribosomal subunit**.
- It is highly lipid-soluble, allowing it to cross the blood-brain barrier effectively, but its use is limited by serious side effects like **Gray Baby Syndrome** and bone marrow suppression.
- **Gray Baby Syndrome** a potentially fatal, toxic reaction to the antibiotic chloramphenicol in neonates (**under 1 month**) and premature infants, characterized by ash-gray skin, cyanosis, abdominal distension, and **cardiovascular collapse**. It results from a **lack of liver enzymes to metabolize the drug, causing high serum levels**.

DNA inhibitors antibiotics

Rifampicin

- potent bactericidal antibiotic that inhibits bacterial DNA-dependent **RNA polymerase**, preventing the synthesis of messenger RNA.
- It is a cornerstone of **tuberculosis (TB) treatment** and is well-known for causing a harmless orange-red discoloration of sweat, tears, and urine.

Quinolones

Table: Pharmacokinetics & Pharmacodynamics of Quinolones

Feature	Quinolones (1st Gen)	Fluoroquinolones (2nd–4th Gen)
Examples	Nalidixic acid, Cinoxacin	Ciprofloxacin (2nd), Levofloxacin (3rd), Moxifloxacin (4th)
Mechanism (PD)	Inhibits DNA Gyrase (Topoisomerase II).	Inhibits DNA Gyrase (Gram-) and Topoisomerase IV (Gram+).
Effect	Bactericidal; prevents DNA supercoiling.	Bactericidal; causes double-stranded DNA breaks.
Absorption (PK)	Moderate oral absorption.	Excellent (70–95%+); high bioavailability.
Distribution (PK)	Poor; restricted to the urinary tract.	Widespread ; high levels in bone, prostate, and lungs.
Metabolism (PK)	Hepatic (partially).	Varies; Moxifloxacin is heavily hepatic.
Excretion (PK)	Renal (Rapidly excreted in urine).	Primarily Renal (Cipro/Levo) or Biliary (Moxi).
Spectrum	Narrow (Gram-negative only).	Broad (Gram-, Gram+, Atypicals, and Anaerobes).

Mechanism of Action of Fluoroquinolones

Metronidazole mechanism of action

Metronidazole enters bacterial cells via passive diffusion.



In anaerobic bacteria, metronidazole is reduced to produce a nitroso free radical.



This free radical binds to DNA, reducing synthesis and causing widespread DNA damage and death.

- **Tinidazole** has a **longer half-life**, requires fewer doses, and is usually better tolerated than metronidazole, but both have similar antimicrobial activity.
- Tinidazole side effects being a metallic/bitter taste in the mouth, nausea, diarrhea, stomach cramps, and headache